

Plugging into the Future: An Exploration of Electricity Consumption Patterns

TEAM ID	SWUID2026202796
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	3 MARKS

6.3 – Data Preparation

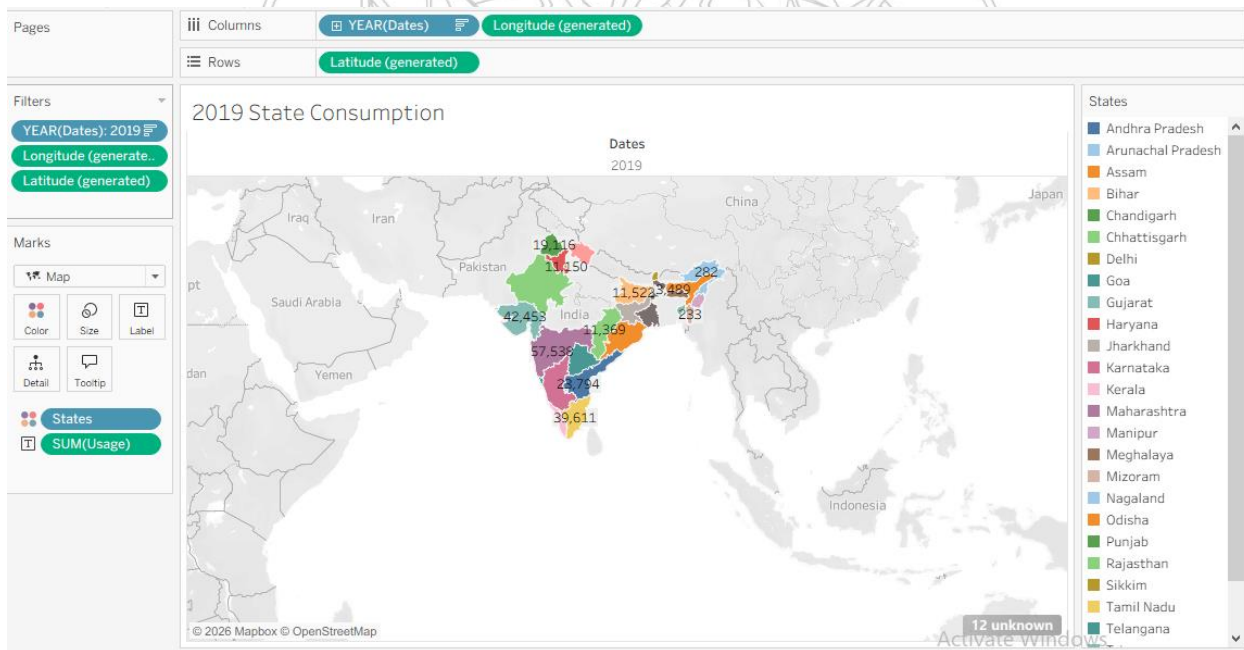
CHART NAME	CHART TYPE
2019 State Consumption	Map chart
2020 State Consumption	Map chart
Total Consumption	Bar chart (column type)
Usage By Region	Side by sidebar chart
Top N and Bottom N	Bar chart
2019- and 2020-month wise consumption	Histogram
Total Consumption Region Wise	Pie chart
Usage Before and After Lockdown	Side by side column chart
Region wise State Usage	Horizontal bar chart
Quarter Usage	Bar chart
Metro city State usage	Symbol map
Usage by year	Line chart

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SHEET -1: 2019 State Consumption

1. Open a New Worksheet.
2. Rename it 2019 State Consumption
3. Drag States to the worksheet.
4. Tableau creates Longitude (generated) and Latitude (generated) automatically.
5. Change Marks type to Map (Filled Map).
6. Drag Usage to Label.
7. Drag States to Colour
8. Drag Dates to Filters and select Year2015.
9. Click Label Show Mark Labels
10. Edit the title as 2019 State Consumption and format the map of needed

OUT PUT:

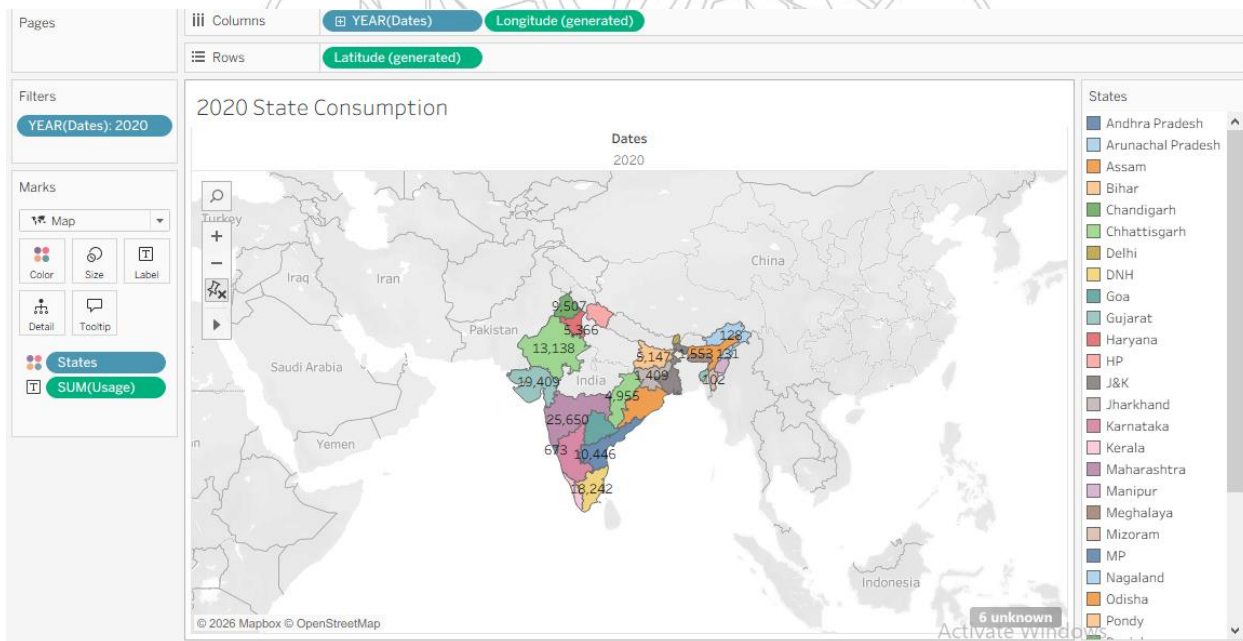


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SHEET -2: 2020 State Consumption

1. Open a New Worksheet.
 2. Rename it 2020 State Consumption
 3. Drag States to the worksheet.
 4. Tableau creates Longitude (generated) and Latitude (generated) automatically.
 5. Change Marks type to Map (Filled Map).
 6. Drag Usage to Label.
 7. Drag States to Colour.
 8. Drag Dates to Filters and select Year2015.
- Chick Label Show Mark Labels
10. Edit the title as 2020 State Consumption and format the map of needed

OUTPUT:

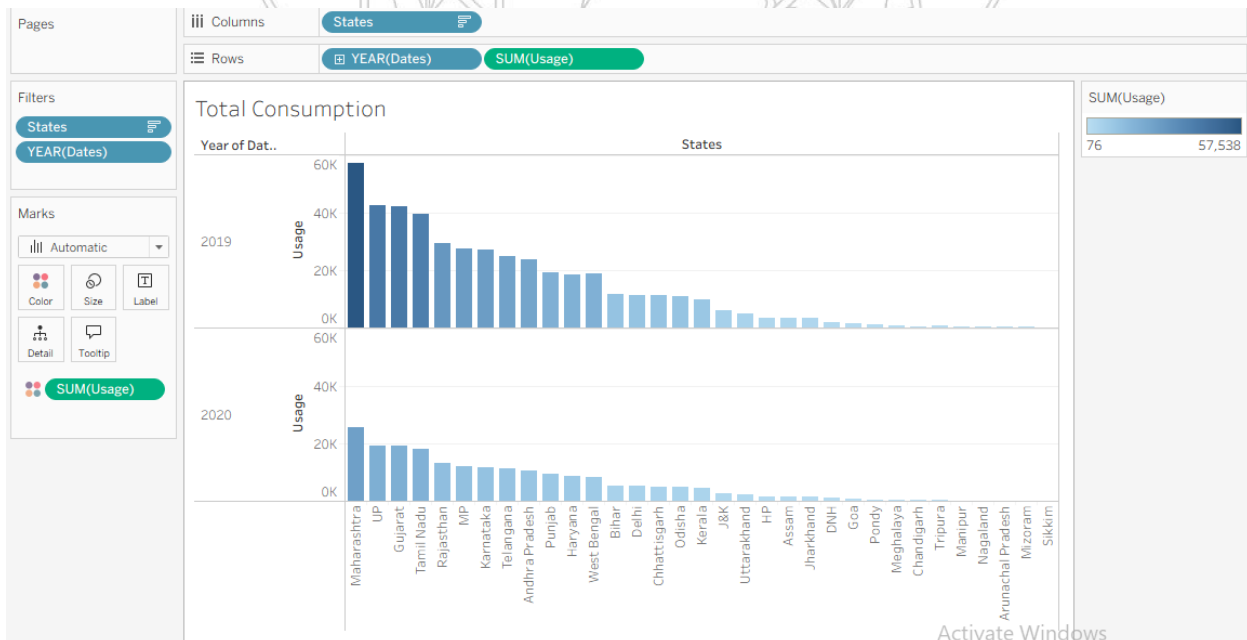


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SHEET -3: Total Consumption

1. Open Tableau and connect to your dataset.
2. Drag States → Columns shelf.
3. Drag Dates → Rows shelf.
4. Click Dates and select YEAR.
5. Drag Usage → Rows beside YEAR(Dates).
5. In Marks, choose Bar chart.
6. Drag SUM(Usage) → Colour to apply colour intensity.
7. Drag States → Filters and select required states.
8. Drag YEAR(Dates) → Filters and choose years (2019, 2020).
9. Sort Usage (Descending) and edit title to Total Consumption for a clean view.

Output:

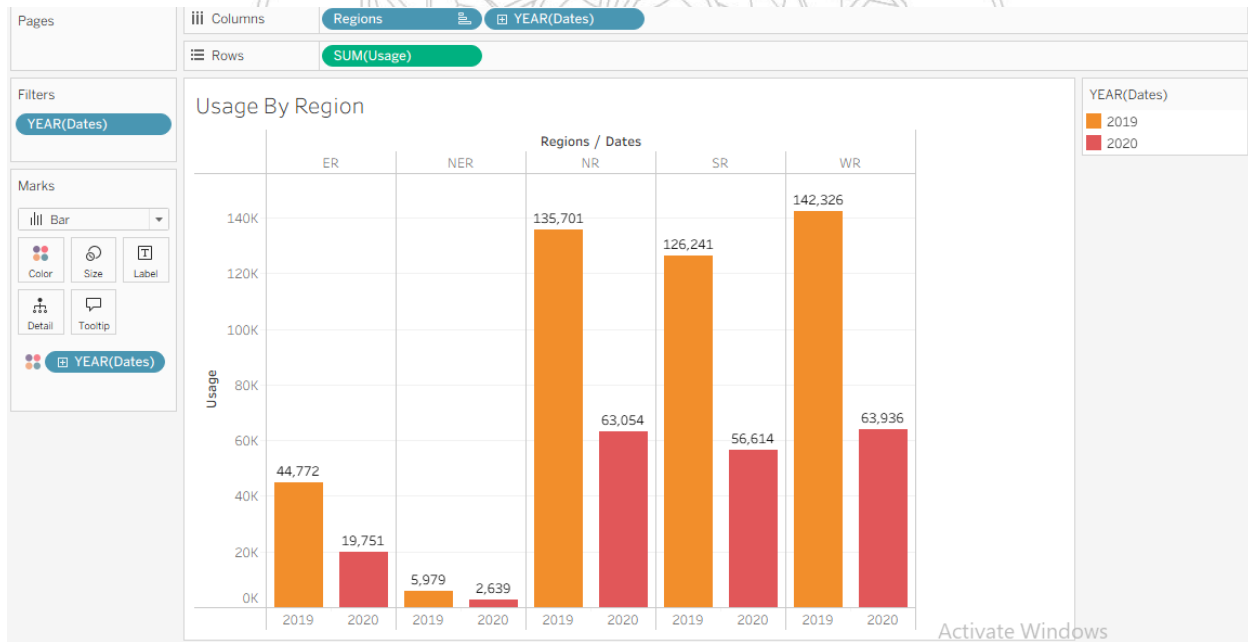


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SHEET -4: Usage by Region

1. Drag Regions → Columns shelf.
2. Drag Dates → Columns beside Regions.
3. Change Dates to YEAR(Dates).
4. Drag Usage → Rows shelf.
5. Select Bar from the Marks card.
6. Drag YEAR(Dates) → Colour to compare years.
7. Drag YEAR(Dates) → Filters and select required years (2019, 2020).
8. Click Label → Show Mark Labels to display usage values.
9. Edit chart title to Usage By Region and format colours/axes for readability.

Output:

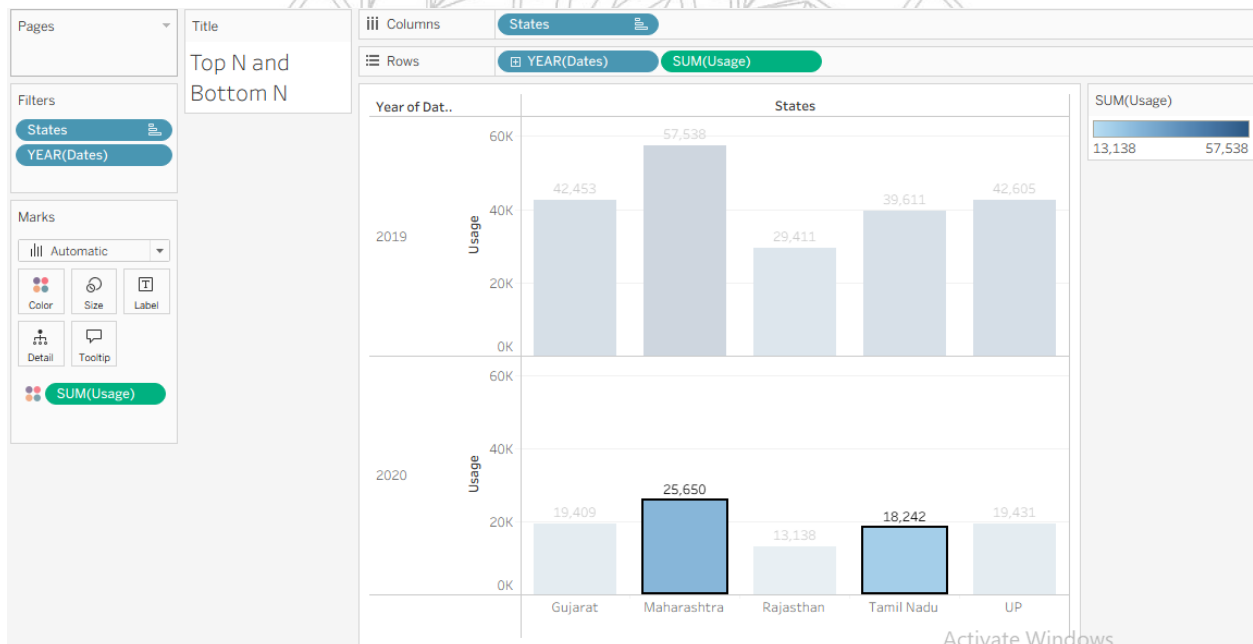


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SHEET -5: Top N and Bottom N

1. Drag States → Columns shelf.
2. Drag Dates → Rows and change to YEAR(Dates).
3. Drag Usage → Rows beside YEAR(Dates).
4. Select Bar chart from the Marks card.
5. Create a Top N / Bottom N filter on States (Filter → Top → By Field → SUM(Usage)).
6. Choose required Top and Bottom values (example: Top 5 & Bottom 5).
7. Drag SUM(Usage) → Colour to highlight usage differences.
8. Enable Show Mark Labels to display usage values on bars.
9. Add YEAR(Dates) and States to Filters

OUTPUT:

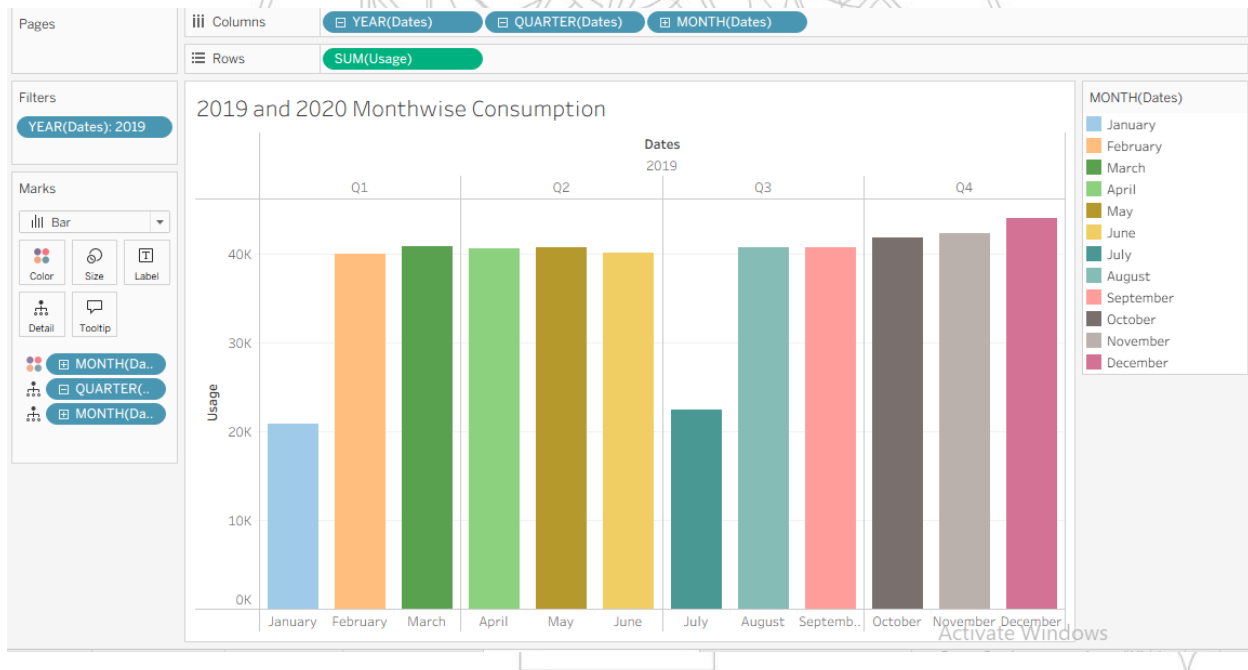


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SHEET-6: 2019- and 2020-Month wise Consumption

1. Drag Dates → Columns shelf.
2. Change Dates hierarchy to YEAR → QUARTER → MONTH.
3. Drag Usage → Rows shelf.
4. Select Bar chart from the Marks card.
5. Drag MONTH(Dates) → Colour to assign different colours for each month.
6. Drag YEAR(Dates) → Filters and select 2019 and 2020.
7. Arrange MONTH(Dates) in chronological order (January–December).
8. Enable Show Mark Labels if values need to appear on bars.
9. Edit the title to 2019- and 2020-Month wise Consumption and format axes/legend for better readability.

OUTPUT:

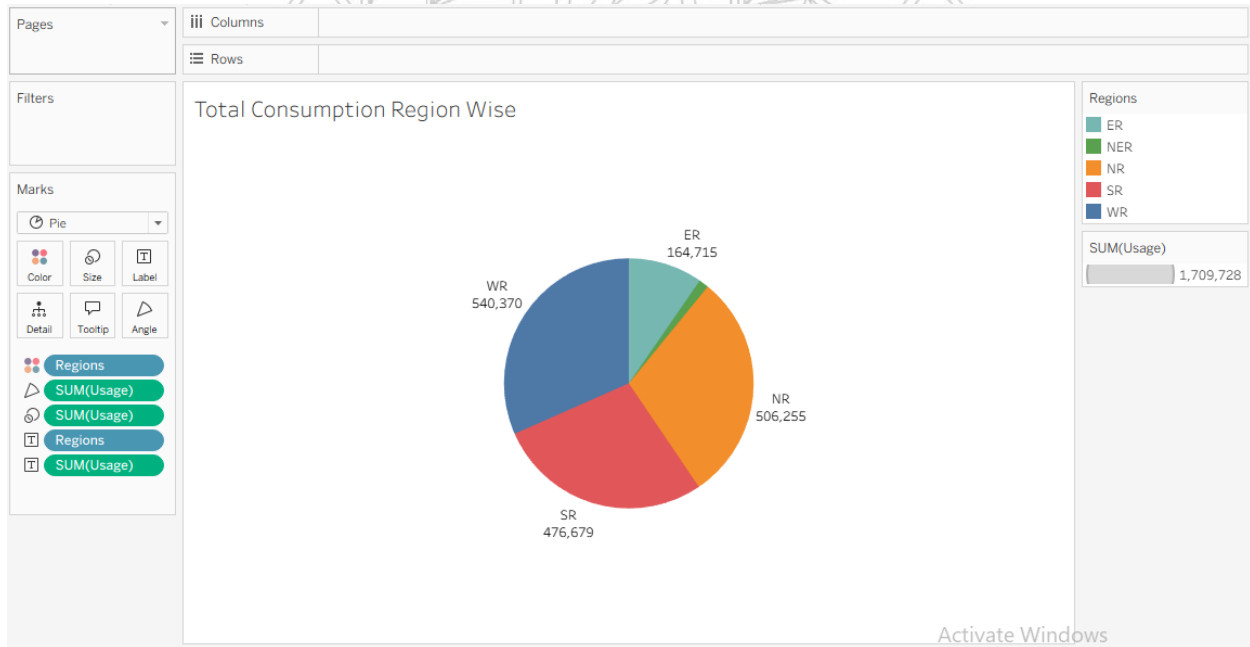


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SHEET -7: Total Consumption Region Wise

1. Open a new worksheet.
2. From Marks, change chart type to Pie.
3. Drag Regions → Colour on the Marks card.
4. Drag SUM(Usage) → Angle to create slice sizes.
5. Drag SUM(Usage) → Size to adjust pie chart size.
6. Drag Regions → Label to display region names.
7. Drag SUM(Usage) → Label to show consumption values.
8. Format labels and colours for clear visualization.
9. Edit the title to Total Consumption Region Wise and fit the chart to the view.

OUTPUT:



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SHEET -8: Usage Before and After Lockdown

1. Open a new worksheet.
2. Drag Lockdown period → Columns shelf.
3. Drag Usage → Rows shelf.
4. Ensure aggregation is SUM(Usage).
5. Select Bar chart from the Marks card.
6. Drag Lockdown period → colour to differentiate categories.
7. Drag SUM(Usage) → Label and enable Show Mark Labels.
8. Format axis labels, colours, and bar spacing for readability.
9. Rename the worksheet title to Usage Before and After Lockdown and fit the chart to the view.

Output:

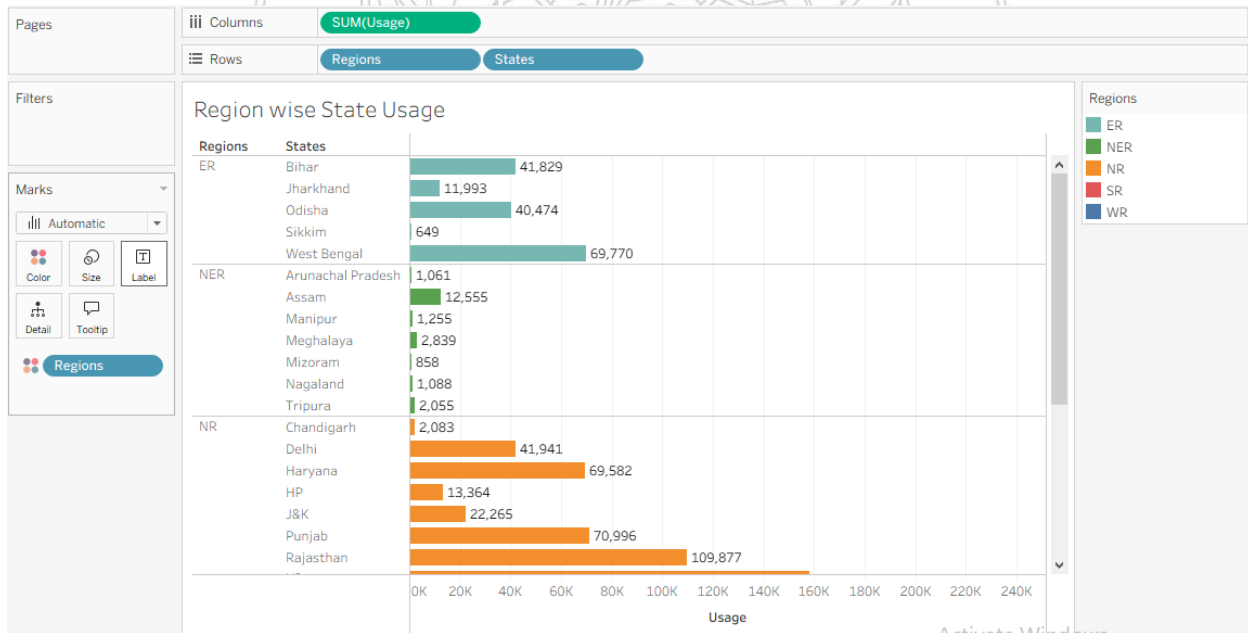


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SHEET -9: Region wise State Usage

1. Drag Regions → Rows shelf.
2. Drag States → Rows beside Regions.
3. Drag Usage → Columns shelf.
4. Ensure aggregation is SUM(Usage).
5. Select Bar chart from the Marks card.
6. Drag Regions → colour to assign different colours by region.
7. Drag SUM(Usage) → Label and enable Show Mark Labels.
8. Sort states in descending order of Usage for better comparison.
9. Edit the worksheet title to Region wise State Usage and fit the view for clear display.

OUPUT:



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SHEET-10: Quarter Usage

1. Drag Dates → Columns shelf.
2. Right-click Dates → Change to Year.
3. Drag Dates again → Columns beside Year.
4. Right-click second Dates → Change to Quarter.
5. Drag Usage → Rows shelf.
6. Ensure aggregation is SUM(Usage).
7. Select Bar chart from the Marks card.
8. Drag SUM(Usage) → Color to apply gradient by usage.
9. Drag SUM(Usage) → Label and enable Show Mark Labels.
10. Sort quarters in chronological order Q1 to Q4 for better comparison.

Output:

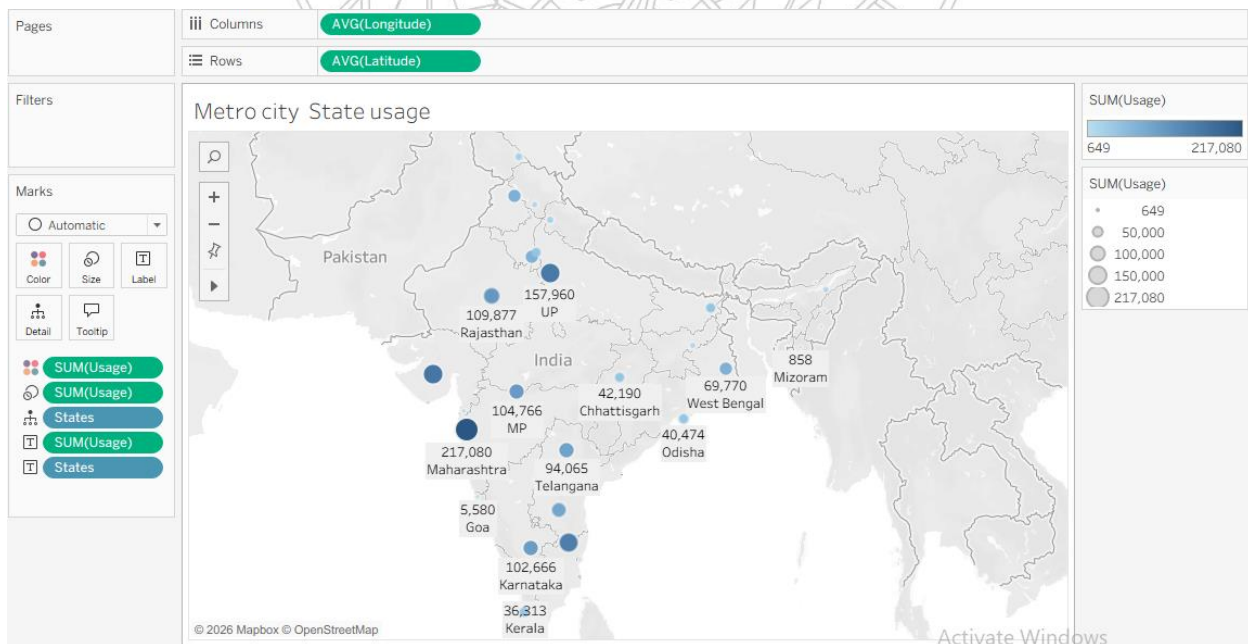


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SHEET -11: Metro city State usage

1. Drag Dates → Columns shelf and change to Year.
2. Drag Dates again → Columns shelf and change to Month.
3. Drag Usage → Rows shelf and set aggregation to SUM.
4. Change Marks card type to Line.
5. Drag YEAR(Dates) → Color to split 2019 vs 2020.
6. Drag SUM(Usage) → Label and set to show Min/Max only.
9. Edit worksheet title to Usage by Year.
10. Click Fit → Entire View for clean layout.

Output:



— BUILDERS —

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SHEET-12: usage by year

1. Drag Dates → Columns shelf.
2. Right-click Dates → Change to Year.
3. Drag Dates again → Columns beside Year.
4. Right-click second `Dates` → Change to Month.
5. Drag Usage → Rows shelf.
6. Ensure aggregation is SUM(Usage).
7. Select Line chart from the Marks card.
8. Drag YEAR(Dates) → Color to separate years.
9. Drag SUM(Usage) → Label and set Marks to Label: Min/Max.
10. Right-click Usage axis → Edit Axis → Fixed range from 0 to 50,000.
11. Edit the worksheet title to Usage by Year and fit the view to Entire View.

Output:

